

Indian TTS Dataset – Build Report

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1 Assignment Details

- **Assignment:** Sarvam AI – TTS Training Dataset (60 min)
- **Dataset:**

<https://huggingface.co/datasets/Kalpeshkhare7777/indian-tts-dataset-marathi-english>

2 What I Built

I curated a balanced, bilingual TTS training dataset of approximately 60 minutes of clean, single-speaker audio split evenly between two Indian languages: Marathi and Indian English. The dataset contains accurate transcripts matched with emotion tags and is published as a public HuggingFace dataset.

- Total duration: approx 60 minutes
- Marathi audio: approx 30 minutes – audiobook (*Shyamchi Aai* by Sane Guruji)
- Indian English audio: approx 30 minutes – audiobook (*The Palace of Illusions* by Chitra Divakaruni)
- Emotion labeling: Sarvam `sarvam-30b-11m` API
- Transcription (Marathi and English): Manual verification against the original book text
- Audio format: WAV, 22050 Hz, mono

3 How the Pipeline Works

The core extraction and tagging pipeline follows the same robust procedure for both the Marathi and Indian English tracks.

3.1 Marathi Track

<https://www.youtube.com/watch?v=SXfzVU20oSE>

Source: *Shyamchi Aai* by Sane Guruji – a classic Marathi literary work with strong emotional range, single-speaker narration, consistent recording quality, and public-domain status.

1. Downloaded the full audiobook audio.
2. Manually identified natural paragraph boundaries by listening and avoided automatic VAD to prevent mid-sentence cuts.

3. Split audio into segments at paragraph breaks, targeting 30–90 seconds per segment.
4. Verified each transcript word-by-word against the printed text of the book.
5. Sent each transcript to **sarvam-30b-11m** with a structured prompt to assign one of five emotion labels.
6. Listened back to each segment to spot-check transcript accuracy and emotion plausibility.

3.2 Indian English Track

https://www.youtube.com/watch?v=TlLyX_tHTrc

Source: *The Palace of Illusions* by Chitra Divakaruni – an English literary work featuring a clear, single-speaker Indian English delivery with natural expressive speech.

1. Downloaded the audiobook audio.
2. Manually identified natural paragraph boundaries by listening to prevent mid-sentence cuts.
3. Split audio into segments at paragraph breaks, targeting 30–90 seconds per segment.
4. Verified each transcript word-by-word against the printed text of the book.
5. Sent each transcript to **sarvam-30b-11m** with a structured prompt to assign one of five emotion labels.
6. Listened back to each segment to spot-check transcript accuracy and emotion plausibility.

3.3 Emotion Tagging Prompt Design

The LLM prompt asked the model to classify each segment transcript into one of five labels:

- emotional
- narrative_calm
- conversational
- sad
- neutral
- tense

The prompt included the transcript text and a brief description of the literary context. This text-based approach is transparent about its limitation: it labels the semantic content of speech rather than its acoustic delivery. This limitation is documented in the dataset README.

3.4 Quality Filtering

- Excluded segments shorter than 20 seconds.
- Changed words from book that differed in audio.

4 Iterations to Improve Quality

4.1 Iteration 1 – Automatic to Manual Splits

Initially attempted VAD-based automatic segmentation. Automatic splits frequently cut mid-sentence during short pauses between clauses, producing grammatically incomplete segments. Switched to fully manual paragraph-level splitting for both tracks. Although slower, this produced clean and complete utterances suitable for TTS training.

4.2 Iteration 2 – Emotion Label Refinement

The first pass used only three emotion labels:

- positive
- negative
- neutral

After reviewing the data, this was found to be too coarse. The labeling scheme was expanded to five labels and all segments were re-tagged. The `narrative_calm` category was introduced specifically for literary audiobook narration.

4.3 Iteration 3 – Transcript Verification

Moved away from relying on raw automated ASR outputs. For both Marathi and Indian English, transcripts were completely cross-checked and verified manually against the official book texts, resulting in near-perfect phonetic alignment and grammar verification.

5 What Worked and What Didn't

Worked Well

- Manual paragraph-level segmentation produced the cleanest and most natural boundaries.
- Established audiobooks provided excellent single-speaker continuity and clean recording environments.
- Sarvam `sarvam-30b-11m` emotion labels were generally consistent and sensible for literary content.

Limitations

- Automatic VAD segmentation frequently produced mid-sentence cuts and required manual rework.
- Emotion tagging based solely on text misses prosodic information. A segment labeled as “neutral” based on text may still be delivered with strong acoustic emotion.

6 Quality Observations and Decisions

The most important quality decision was selecting source material deliberately rather than scraping content at scale.

Both *Shyamchi Aai* and *The Palace of Illusions* were chosen because they provide:

- Single-speaker narration
- Consistent recording quality throughout
- Emotionally varied content
- Clean literary diction suitable for TTS training

A key quality indicator for TTS data is whether a model trained on the dataset would generate natural-sounding speech. The collected segments are complete, exhibit natural tone, and maintain a consistent speaker identity throughout. These characteristics are valuable for single-speaker TTS fine-tuning.

7 What I Would Improve Given More Time

- Add acoustic-based emotion tagging using pitch, energy, and tempo features to complement text-based labels.
- Include speaker metadata such as gender, approximate age, and dialect region.

8 Tools and APIs Used

- Sarvam LLM (`sarvam-30b-11m`) – emotion/style classification
- `yt-dlp` – YouTube audio download
- FFmpeg – audio conversion, segmentation, and processing
- Python – pipeline scripting (`datasets`, `requests`, `pathlib`)
- HuggingFace Datasets – dataset packaging and publishing
- Claude (Anthropic) – pipeline design, code review, and README writing